

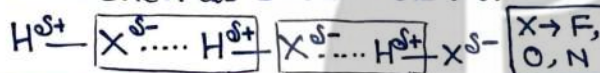
Thermodynamics

* Hydrogen Bonding

The attractive force which binds hydrogen atom of one molecule with the highly electronegative atom (F, O or N) of the same or another molecule called Hydrogen bonding. It is represented by a dotted line.

• Cause of Formation of Hydrogen Bond
F hydrogen is bonded to highly electronegative element X, the shared pair of electrons moves far away from hydrogen atom.

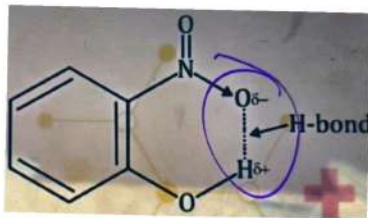
Since, there is displacement of electrons towards X, the hydrogen acquires partial positive charge (δ^+), while X gains partial negative charge (δ^-). It results in the formation of a polar molecule with electrostatic force of attraction as shown below.



* Type of H-Bonds $\rightarrow$ 2 types

a) Intermolecular Hydrogen Bond

Such bond exist between two different molecules of the same or different compounds, eg. Hydrogen bond in case of HF molecules, alcohol or water etc. It results in association of two molecules and hence results in increased melting points, boiling points and solubility



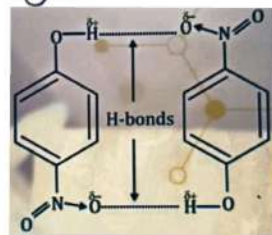
b) Intramolecular Hydrogen Bond

Such bond exists when hydrogen atom is in between two highly electronegative (F, O, N) atom present within the same molecule. eg. in o-nitrophenol, the hydrogen is in between the two oxygen atoms.

This bonding prevent the association of molecules, making the molecules more sechuled in comparison

of molecules where intermolecular hydrogen bonding occurs. Hence, it decreases melting point, boiling point, and solubility.

eg. o-nitrophenol has lower boiling point than p-nitrophenol.



* Concept of System & Surrounding

• **Thermodynamics**:- It is the study of various forms of energies and their inter-relationship

• **System**:- Part of universe in which we are interested

• **Surroundings**:- It is also the part of universe other than system

• Universe is composed of system & surrounding

$$\text{Universe} = \text{System} + \text{Surrounding}$$

* Types of System

1) **Open System**:- A system is said to be open when it exchanges both matter and energy with the surrounding. e.g. $\rightarrow$ Water placed in open vessel, exchanges matter as well as energy with surrounding.

2) **Closed System**:- A system is said to be closed when it exchange only energy with surrounding. e.g. If Close the container of water, then it only exchanges energy but not the matter with the surrounding.

3) **Isolated System**:- A system is said to be isolated when it neither exchanging matter nor energy with surround.

e.g. $\rightarrow$ Water kept in thermos.

* Extensive and Intensive Properties

① **Intensive Property**:- Those properties of system which are independent of amount of substance. e.g. → Density, Refractive index, viscosity, pressure, Temperature, concentration.

② **Extensive Property**:- Those properties which depends upon the amount of substance. e.g. → Mass, mole, volume, enthalpy, Gibbs free energy, heat capacity, Entropy Work. etc.

□ If we take ratio of two extensive properties then it become intensive properties. e.g. $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$

* State & Path Function

• **State of System**:- It is simply the condition in which a system is present. Condition is express in term of temp, pressure and volume.

• **State Function**:- Those properties of a system that depends upon initial and final state of system. These properties do not depend upon path followed. e.g. → P, V, T, Internal energy, Gibbs free energy, Entropy.

• **Path Function**:- Those properties of a system that depends upon the path followed by the system. e.g. → Work, heat, heat capacity are path function.

* Types of Processes

The state of variable can be changed via thermodynamic process. These process are following types:-

① **Adiabatic Process**: In adiabatic process the system does not exchange heat with its surrounding i.e. no heat can flow in or out of the system.

• For adiabatic process, $dq = 0$

② **Isothermal Process**:- The isothermal process is carried out at constant temperature i.e. it is the process in which temperature remains fixed. • For isothermal process, $dT = 0$

③ **Isobaric Process**:- In isobaric process change of state is brought about at constant pressure i.e. $dp = 0$

④ **Isochoric Process**:- In isochoric process volume of the system remains constant i.e. $dv = 0$

⑤ **Cyclic Process**:- When system is undergoes number of different and finally return to initial state, this process, is called cyclic process. e.g. - in a process shows below the system returns to the original state after successive changes.

$A(P_1, V_1, T_1) \rightarrow B(P_2, V_2, T_2) \rightarrow C(P_3, V_3, T_3) \rightarrow A(P_1, V_1, T_1)$

⑥ **Reversible Process**:- A reversible system is defined as a system where change takes place infinitesimally slow step and direction of which can be reversed at any step by bringing a small change.

⑦ **Irreversible Process**:- An irreversible process is one, the direction of which cannot be reversed by small change in variables. All natural process are generally irreversible and the net work done in irreversible Process is somewhat less than that in reversible process i.e. $W_{irr} < W_{rev}$

* Work (Mechanical/P-V work)

Let a cylinder a gas filled with a friction less piston, when we can apply external pressure on gas Compression takes place from V_i to V_f

• Work done in process can be calculated in two ways.

i) Work done for Irreversible

ii) Work done for reversible process



① Work done for irreversible process

For compression volume change can be written as $= l \times A$ or $\Delta V = V_f - V_i$ and pressure $= \frac{\text{Force}}{\text{Area}}$ and Force $= P_{\text{ext}} \times A$
 $\therefore$ Work done $= \text{Force} \times \text{distance}$
 $= P_{\text{ext}} \times A \times l$
 $= P_{\text{ext}} \times \Delta V$

$$W_{\text{irr}} = \text{Work done} = -P_{\text{ext}} \Delta V$$

② Work done for Reversible Process

As we know work done in reversible process takes place in infinite steps

$$W_{\text{rev}} = - \int_{V_i}^{V_f} P_{\text{ext}} dV$$

For compression $P_{\text{ext}} = P_{\text{int}} + dP$ ($P_{\text{ext}} > P_{\text{int}}$)

For expansion $P_{\text{ext}} = P_{\text{int}} - dP$ ($P_{\text{ext}} < P_{\text{int}}$)

or $P_{\text{ext}} = P_{\text{int}} \pm dP$

$$W_{\text{rev}} = - \int_{V_i}^{V_f} (P_{\text{int}} \pm dP) dV = - \int_{V_i}^{V_f} P_{\text{int}} dV \pm dP dV$$

$$W_{\text{rev}} = - \int_{V_i}^{V_f} P_{\text{int}} dV \quad (\because dP dV = \text{very small})$$

For ideal gas equation $= PV = nRT$

$$\Rightarrow P = \frac{nRT}{V}$$

$$W_{\text{rev}} = - \int_{V_i}^{V_f} \frac{nRT}{V} dV = -nRT \int_{V_i}^{V_f} \frac{dV}{V} = -nRT \ln \frac{V_f}{V_i}$$

$$W_{\text{rev}} = -nRT \ln \frac{V_f}{V_i} \quad \left(\because \int_{V_i}^{V_f} \frac{dV}{V} = \ln \frac{V_f}{V_i} \right)$$

$$W_{\text{rev}} = -2.303 nRT \log \frac{V_f}{V_i} \quad [\ln = 2.303 \log]$$

* Sign convention for Work:-

i) Work done on system $= +ve$ (compression)

We know that $W = -P_{\text{ext}}(V_f - V_i)$

$$\therefore V_f < V_i \Rightarrow = (-ve)(-ve) = +ve$$

ii) Work done by the system (expansion)

$$W = -P_{\text{ext}}(V_f - V_i) = -ve$$

$$\text{For expansion } (V_f > V_i) = (-ve)(+ve) = -ve$$

* Work of free expansion

If the gas expands in vacuum then it is called free expansion. In vacuum, $P_{\text{ext}} = 0$

$$W = -P_{\text{ext}} \Delta V = 0 \quad \boxed{W = 0}$$

* Heat

It is form of energy that transfer from one body to another body when their temperature is different

It is denoted by 'q'

* Heat given by the system, $q = -ve$

* Heat absorbed by the system, $q = +ve$

* Heat is a path function

* Extensive properties

* Unit is Joule or Calorie.

* Heat Capacity

It is defined as heat required to raise the temperature of system by one degree (Celsius or Kelvin)

It is denoted by 'C'

Mathematically,

$$C = \frac{q}{\Delta T} \quad \text{Unit} = \text{JK}^{-1}$$

* It is a path function

* It is an extensive property

* Specific Heat capacity

It is defined by heat required to raise the temperature of 1g substance by one degree ($^{\circ}\text{C}/\text{K}$)

It is denoted by C_s

Mathematically,

$$C_s = \frac{q}{\Delta T_m} \text{ or } \frac{C}{m} \quad \therefore \text{Unit} = \text{JK}^{-1}\text{g}^{-1}$$

* State function * Intensive property

* Molar Heat Capacity

It is defined as heat required to raise the temperature of 1 mole of substance by 1 degree ($^{\circ}\text{C}/\text{K}$)

It is denoted by C_m

Mathematically,

$$C_m = \frac{q}{\Delta T n} \text{ or } \frac{C}{n} \quad \text{Unit} = \text{J K}^{-1} \text{mol}^{-1}$$

* State function * Intensive property

* Internal Energy

It is defined as sum total of all the forms of energy of a system.

• It is denoted by U or E .

* $U = \text{Kinetic Energy} + \text{Potential energy} + \text{Vibrational energy} + \dots$

• In thermodynamics we usually talk changes of energy i.e. we talk about ΔU or ΔE .

* Internal energy is state function.

* Let us take a chemical Reaction



If U_R is internal energy of reactant & U_P is internal energy of Product

then $\Delta U = U_P - U_R$

• If $U_P > U_R$, then $\Delta U = +ve$

• If $U_R > U_P$, then $\Delta U = -ve$

Therefore value of ΔU depends upon initial & final state of reaction hence internal energy is state function & Extensive Properties

* First Law of Thermodynamics

This law is also known as law of conservation of energy. This law states that "Energy can neither be created nor be destroyed, only it can be transferred from one form to another." OR,

"Total energy of the isolated system is constant."

* Examples

① Photosynthesis $\rightarrow$ Light energy converted into chemical energy

② Burning of Wood $\rightarrow$ Chemical energy converted into light and heat energy

③ Electrical Bulb $\rightarrow$ Electrical energy converted into light energy.

* Derivation:

Suppose a system having internal energy U_1 , if we wanted to change the internal energy, we either have to supply the heat or can do work on the system.

$$U_1 \xrightarrow{q} U_1 + q \xrightarrow{w} U_1 + q + w$$

(final $\rightarrow U_2$)

Let U_1 is the initial internal energy and Let supplying heat & doing work internal energy changes to $U_1 + q + w$ consider this energy as U_2

$$U_2 = U_1 + q + w \text{ or } U_2 - U_1 = q + w$$

$$\Delta U = q + w$$

or it can be written as

$$\Delta U = q + (-P\Delta V) \text{ or } \Delta U = q - P\Delta V$$

$$\text{or } q = \Delta U + P\Delta V$$

$\hookrightarrow$ derivation of 1st Law of thermodynamics.

* Enthalpy (H)

"It is defined as part of internal energy that can be converted into heat"

• It is denoted as H

• Extensive property & State function

• Mathematically, $H = U + PV$ — ①

• According to 1st law $q = \Delta U + P\Delta V$ — ②

• If the 'q' heat is absorbed or released at constant pressure by a system, $q_p = \Delta U + P\Delta V$ — ②

$$\text{or } q_p = U_2 - U_1 + P(V_2 - V_1)$$

$$= U_2 - U_1 + PV_2 - PV_1$$

$$= U_2 + PV_2 - U_1 - PV_1$$

$$q_p = U_2 + PV_2 - (U_1 + PV_1)$$

• From 1st $q_p = H_2 - H_1$

$$q_p = \Delta H$$

Put this value in 2nd, we get

$$\Delta H = \Delta U + P\Delta V$$

• Heat absorbed or released at constant volume

$$q_v = \Delta U + P\Delta V$$

at constant volume $\Delta V = 0$

$$q_v = \Delta U$$

Heat absorbed or released at constant volume is equal to the change in internal energy.

(NB) $q_p = \Delta H$ $q_v = \Delta U$

* Relation between q_p & q_v

We know that $\Delta H = \Delta U + P\Delta V$

$$\text{or } \Delta H = \Delta U + P(V_2 - V_1) = \Delta U + PV_2 - PV_1$$

• According to ideal gas equation,

$$PV = nRT$$

$$PV_2 = n_2 RT \quad \& \quad PV_1 = n_1 RT$$

$$\Delta H = \Delta U + n_2 RT - n_1 RT$$
$$= \Delta U + (n_2 - n_1) RT$$

$$\Delta H = \Delta U + \Delta n_g RT$$

Δn_g is the change in no. of moles in a chemical reaction in gaseous state.

OR $q_p = q_v + \Delta n_g RT$

* C_p and C_v

① Heat Capacity at constant Pressure (C_p)

We know that heat capacity (C)

$$C = \frac{q}{\Delta T} \quad \& \quad C_p = \frac{q_p}{\Delta T}$$

$$\therefore q_p = C_p \Delta T$$

OR $\Delta H = C_p \Delta T$ ($\because q_p = \Delta H$)

② Heat Capacity at constant Volume

We know that, heat capacity (C_v)

$$C = \frac{q}{\Delta T} \quad \text{or} \quad C_v = \frac{q_v}{\Delta T}$$

$$q_v = C_v \Delta T$$

OR $\Delta U = C_v \Delta T$ ($\because q_v = \Delta U$)

* Relation between C_p & C_v for one mole of ideal gas.

We know that, $H = U + PV$

$$\text{or } \Delta H = \Delta U + \Delta PV$$

and for 1 mole of ideal gas $PV = RT$

$$\Delta H = \Delta U + \Delta RT$$

$$\text{OR } C_p \Delta T = C_v \Delta T + R \Delta T \quad \left[\because \begin{matrix} \Delta H = C_p \Delta T \\ \Delta U = C_v \Delta T \end{matrix} \right]$$

dividing both side by ΔT , we get

$$\text{OR } \frac{C_p \Delta T}{\Delta T} = \frac{C_v \Delta T}{\Delta T} + \frac{R \Delta T}{\Delta T}$$

$$\text{OR } C_p = C_v + R$$

$$\text{OR } C_p - C_v = R$$

* Measurement of Enthalpy and Internal Energy Change.

ΔH & ΔU is calculated with the help of a experimental techniques called as calorimetry.

@ Calculation of ΔU :-

The experimental set up or method which is used to calculate internally energy change is called Bomb Calorimeter.

• **Apparatus** $\rightarrow$ It is consist of very strong steel or bomb calorimeter for sample. Water is present outside of the bomb in another container. In outside container, thermometer and stirrer is present.

• **Process** :- A sample of given mass placed in a bomb, and it is burn

with the help of firing lead. And change in temperature is measured (ΔT). Therefore heat capacity (C) of calorimeter is easily calculated and ΔU can be calculated by following equation.

$$\Delta U = \frac{C \Delta T M}{m}$$

C = heat capacity

ΔT = change in temp

M = Molar mass

m = Given mass.

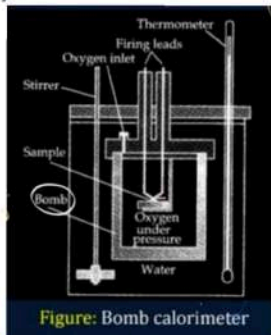


Figure: Bomb calorimeter

(b) Calculation of ΔH

ΔH is calculated with the help of coffee cup calorimeter.

A thermometer is placed in coffee cup and stirrer is also attached. Reactant are poured inside the cup and reaction is allowed to take place. If this reaction is endothermic then q is absorbed & if exothermic then q is released. The ΔH is calculated with the help of

$$\Delta H = C \Delta T \times C_s$$



* Types of Enthalpy

* Reaction enthalpy

It is defined as an enthalpy change when no. of moles of reactant react to produce product.



• It can be +ve i.e. endothermic rxn

• It can be -ve i.e. exothermic reaction

* $\Delta_r H$ = Sum of Enthalpies of reactants.

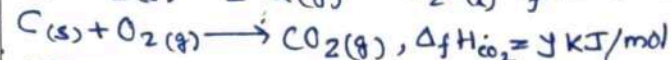
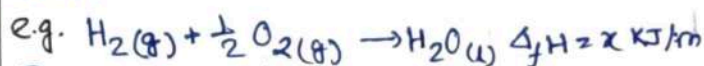
* Standard Enthalpy of Reaction

* It is defined as enthalpy change when no. of moles of reactants react to produce product, but all

reactants & products must be in their standard state. (Most stable state). It is denoted as $\Delta_r H^\circ$

* Standard Enthalpy of Formation

It is the change in enthalpy when one mole of a substance is formed from their constituent element present in standard state.

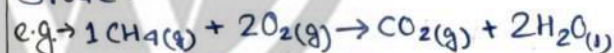


• From sign convention, enthalpy of formation of element present in their standard state is taken as zero. eg. C (graphite) is solid, Br is liquid, I is solid, H is gas.

* Enthalpy for different types of Reactions

* Standard Enthalpy of Combustion

It is defined as enthalpy ($\Delta_c H$) change when one mole of substance is completely burnt in presence of oxygen, provided all reactants and product must be in their standard state.

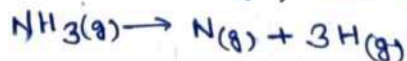
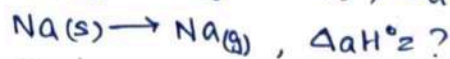
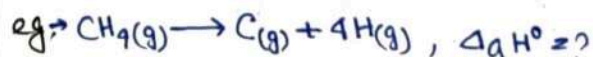


• Calorific value:- It is the heat produced or absorbed when 1gm of any substance is completely burnt in O_2 .

$$\text{Calorific value} = \frac{\Delta_c H}{m}$$

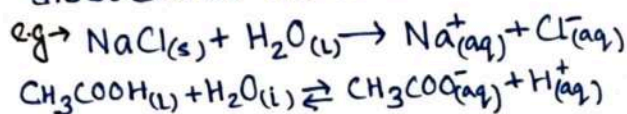
* Standard Enthalpy of Atomisation

It is change in enthalpy ($\Delta_a H^\circ$) when one mole of substance is completely dissociated into gaseous atoms.



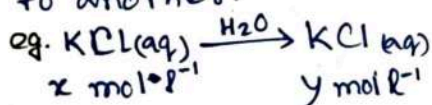
* Enthalpy of ionisation, $\Delta_{\text{ion}} H^\ominus$

It is the change in enthalpy when one mole of electrolyte is completely dissociated into ions.



* Enthalpy of dilution, $\Delta_{\text{dil}} H^\ominus$

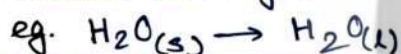
It is defined as enthalpy change when one mole of a substance is diluted from one concentration to another.



* Enthalpy change during Phase Transformations

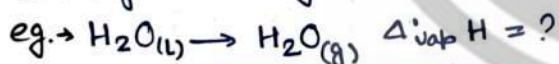
a) Enthalpy of Fusion, $\Delta_{\text{fus}} H^\ominus$

It is defined as enthalpy change when one mole of a solid substance changes into liquid.



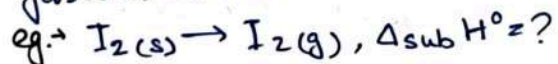
b) Enthalpy of vaporisation

It is defined as enthalpy change when one mole of a liquid substance changes into gaseous substance.



c) Enthalpy of Sublimation

It is defined as enthalpy change when one mole of substance changes directly from solid to gaseous state.



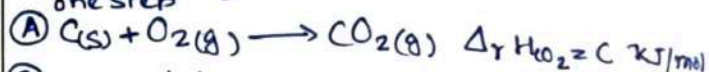
* Hess's Law

- It was given by Russian chemist G. H. Hess.
- It states that "Total enthalpy of reaction remains same, whether it is complete in one step or more than one step."

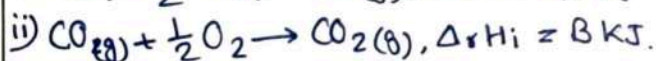
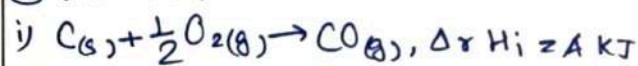
eg. Formation of CO_2 takes place

in two ways —

one step



two step.



$$\Delta_r H_i + \Delta_r H_{ii} = \Delta_r H(\text{CO}_2)$$

$$A + B = C$$

- This is because Enthalpy is a state function; it depends only on initial & final state.

- This law is according to the Law of conservation of energy.

* Application of Hess's Law

*) For determination of —

i) formation of those compounds which cannot be determined experimentally.

ii) enthalpy of extremely slow rxn

iii) enthalpies of transformation of one allotropic form into another.

iv) enthalpy of hydration

v) bond energy

vi) resonance energy

vii) Lattice energy.

* Limitation of First Law of Thermodynamics

1) It doesn't tell us the direction of heat flow. In reality, heat always flows from hot to cold.

2) It doesn't say if a process is spontaneous or possible.

3) It says energy is conserved, but in real life, not all heat can be fully converted to work without some loss.

* Spontaneous & Non spontaneous Process

* Spontaneous Process: A change or process that take place itself or require little initiation.

* Non-spontaneous Process: A change or process that require continuous supply of energy.

* Example of Spontaneous process

[a] Which take place by themselves

i) Dissolution of salt or sugar in water
 $\text{Sugar} + \text{Water} \rightarrow \text{Aqueous solution of sugar.}$

ii) Evaporation of water from water reservoirs such as pond, lakes, river, sea, open vessels. etc.



iii) Flow of heat from hotter to a colder body

iv) Mixing of different gases

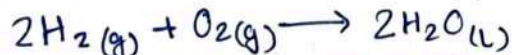
v) Flow of water down hill.

vi) A piece of sodium metal reacts violently with water to form sodium hydroxide and hydrogen gas.



[b] Which takes place on initiation

i) Combination of hydrogen & oxygen to form water (initiated by passing an electric spark)



ii) Reaction between methane and oxygen to form carbon dioxide and water (initiated by ignition)

* Example of Non spontaneous Process

i) Flow of heat from cold body to a hot body

ii) Flow of water from ground to overhead tank

iii) Decomposition of water into hydrogen and oxygen.

iv) Diffusion of a gas from low pressure to high pressure

v) Hydrolysis of sodium chloride

* Entropy

Entropy is the measure of disorderness or randomness of system. More the disorderness of the system more is the entropy.

- Entropy is Extensive property and State Function.

- In case of solid, liquid and Gas, order of entropy $\text{Solid} < \text{Liquid} < \text{Gas}$

- In case of vaporisation or melting entropy increase.

- In case of freezing entropy is decreasing.

- For a reaction, $A \rightarrow B$

$$\Delta S = S_{\text{product}} - S_{\text{reactant}}$$

- If two system are placed at different temperature & some amount of heat is supplied to them, the entropy change for system at lower temperature is more.

- Mathematically, $\Delta S = \frac{q}{T}$

* Entropy change for Reversible Process

i) Consider isothermal reversible process

Let heat is absorbed by the system & heat is released by the surrounding then,

$$\Delta S_{\text{system}} = +\frac{q}{T} \quad \& \quad \Delta S_{\text{surrounding}} = -\frac{q}{T}$$

$$\begin{aligned} \therefore \Delta S_{\text{universe}} &= \Delta S_{\text{syst}} + \Delta S_{\text{sur}} \\ &= +\frac{q}{T} - \frac{q}{T} \\ &= 0 \end{aligned}$$

ii) Consider adiabatic reversible process

In case of adiabatic process is neither absorbed or released i.e. $q=0$

$$\Delta S_{\text{univ}} = 0$$

* Entropy change for irreversible Process

'Consider a surrounding is at higher temperature (T_1) & system is at lower temperature'. Suppose heat is released by surrounding & absorbed by system therefore, $<$

$$\Delta S_{\text{sys}} = \frac{q}{T_2} \quad \& \quad \Delta S_{\text{surr}} = -\frac{q}{T_1}$$

$$\Delta S_{\text{univ}} = \frac{q}{T_2} - \frac{q}{T_1} = q \left(\frac{1}{T_2} - \frac{1}{T_1} \right) = q \left(\frac{T_1 - T_2}{T_1 T_2} \right)$$

As we know, $T_1 > T_2$, then $(T_1 - T_2) = +ve$

$$\Delta S_{\text{univ}} = +ve$$

* Note -

$\Delta S_{\text{univ}} = 0$ (Reversible Process)

$\Delta S_{\text{univ}} = 0$ (Cyclic process)

$\Delta S_{\text{univ}} = +ve$ (Irreversible process)
(Natural process)

* Second Law Thermodynamics

"For any Irreversible process, the total entropy keep on increasing"

* $\Delta S_{\text{Total}} = +ve \rightarrow$ spontaneous pr.

$\Delta S_{\text{Total}} = -ve \rightarrow$ Non-spontaneous process

$\Delta S_{\text{Total}} = 0 \rightarrow$ Equilibrium.

* Gibbs Free Energy

The maximum amount of energy of a system that can be converted into useful work is called free energy or Gibbs free energy.

• State function • Extensive Property

• Mathematically :- $G = H - TS$

* For a isothermal process, Consider

$G_1 = H_1 - TS_1$, for Initial state

$G_2 = H_2 - TS_2$, for final state

$$G_2 - G_1 = (H_2 - TS_2) - (H_1 - TS_1)$$

$$\Delta G = H_2 - TS_2 - H_1 + TS_1$$

$$\Delta G = H_2 - H_1 - TS_2 + TS_1$$

$$\Delta G = (H_2 - H_1) - (TS_2 - TS_1)$$

$$\Delta G = (H_2 - H_1) - T(S_2 - S_1)$$

$$\Delta G = \Delta H - T \Delta S \rightarrow \text{Gibbs-Helmholtz Equation}$$

* Spontaneity of Gibbs Free Energy

For a isothermal process, heat is absorbed by the system & released by surrounding at cont pressure.

$$-q_{\text{surr}} = +q_{\text{sys}} = \Delta H_{\text{sys}}$$

$$\text{or } q_{\text{surr}} = -\Delta H_{\text{sys}}$$

We know that, $\Delta S_{\text{Total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$

$$\Delta S_{\text{Total}} = \Delta S_{\text{sys}} + \frac{q_{\text{surr}}}{T}$$

$$\Delta S_{\text{Total}} = \Delta S_{\text{sys}} - \frac{\Delta H_{\text{sys}}}{T}$$

or By multiplying T on both side

We get, $T \Delta S_{\text{Total}} = T \Delta S_{\text{sys}} - \Delta H_{\text{sys}}$

$$T \Delta S_{\text{Total}} = -\Delta H_{\text{sys}} + T \Delta S_{\text{sys}}$$

$$\text{or } T \Delta S_{\text{Total}} = -(\Delta H_{\text{sys}} - T \Delta S_{\text{sys}})$$

$$\text{or } -T \Delta S = \Delta H_{\text{sys}} - T \Delta S_{\text{sys}}$$

$$-T \Delta S_{\text{Total}} = \Delta G$$

• From this equation, we can conclude
For spontaneous process,

$$\Delta S_T = +ve, \Delta G = -ve$$

For Non spontaneous process,

$$\Delta S_T = -ve, \Delta G = +ve$$

For Equilibrium, $\Delta S_T = 0, \Delta G = 0$

* Equilibrium and Its Classification

Equilibrium represent the state of a process in which the properties like temperature, pressure, concentration of the system do not show any changes with the passage of time.

• The equilibrium may be classified as:

1) Physical Equilibrium:- The equilibrium set up in physical process is called physical equilibrium. For example, equilibrium in melting of ice to water, evaporation of water to steam.

2) Chemical Equilibrium:- The equilibrium set up in chemical process is called chemical equilibrium. For example, equilibrium in decomposition of calcium carbonate, reaction between hydrogen and iodine.